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01286326 THEORY OF COMPUTATION ASSIGNMENT 2

PROBLEM 1

Suppose M is a finite automaton $(Q, \Sigma, \delta, q_0, F)$ where

- $Q = \{q_0, q_1, q_2, q_3, q_4\}$
- $\Sigma = \{0, 1\}$
- δ is described as

	0	1
q_0	q_1	q_3
q_1	q_2	q_1
q_2	q_4	q_1
q_3	q_3	q_4
q_4	q_0	q_2

- q_0 is the initial state,
- $F = \{q_1, q_4\}$

Show how to construct a regular expression α equivalent to this automaton, i.e. $L(\alpha) = L(M)$.

PROBLEM 2

Let β be the following regular expression over the alphabet $\Sigma = \{0, 1\}$:

$$\beta = (01 \cup 011)^+(10 \cup 110)^*$$

Show how to construct a nondeterministic finite automaton N equivalent to this regular expression, i.e. $L(N) = L(\beta)$.

PROBLEM 3

Let R be the following language over the alphabet $\Sigma = \{0, 1\}$:

$$R = L((0 \cup 10)^*) - L((\Sigma\Sigma)^*)$$

Find a regular expression φ such that $L(\varphi) = R$.

PROBLEM 4

For each language below, prove or disprove whether it is regular. Here, $\Sigma = \{0, 1\}$.

- (1) $L_1 = \{w \in \Sigma^* \mid w \text{ is a palindrome}\}$. Note that a *palindrome* is a string that reads the same forward and backward, such as 10101 and 0110.
- (2) $L_2 = \{0^m 1^n \mid m \geq n \geq 1 \text{ and } m \text{ is divisible by } n\}$.
- (3) $L_3 = \{ww \mid w \in \Sigma^*\} \cap \{w \in \Sigma^* \mid w \text{ contains no consecutive 0s and no consecutive 1s}\}$.

PROBLEM 1

Suppose M is a finite automaton $(Q, \Sigma, \delta, q_0, F)$ where

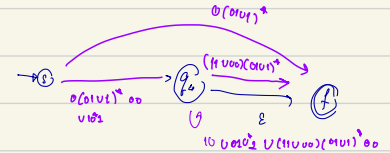
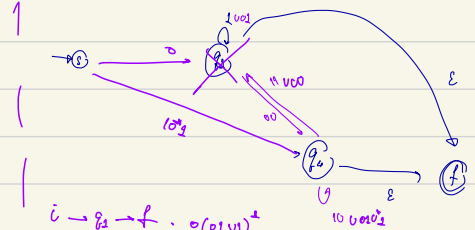
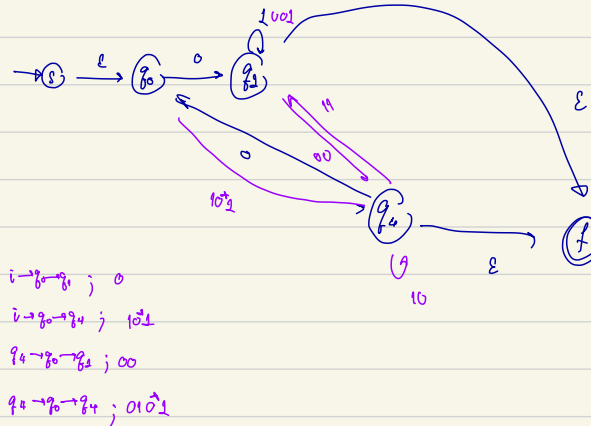
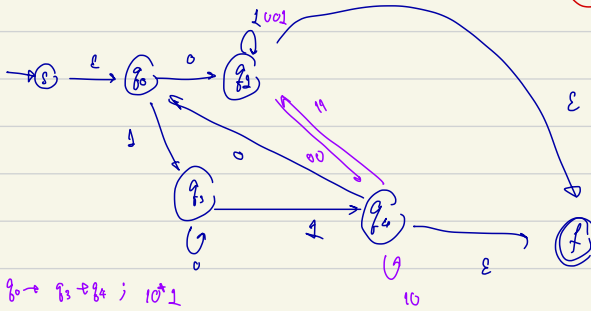
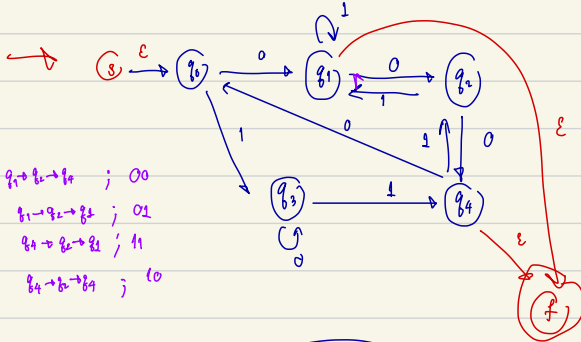
teacher's solution.

- $Q = \{q_0, q_1, q_2, q_3, q_4\}$
- $\Sigma = \{0, 1\}$
- δ is described as

	0	1
q_0	q_1	q_3
q_1	q_2	q_1
q_2	q_4	q_1
q_3	q_3	q_4
q_4	q_0	q_2

- q_0 is the initial state,
- $F = \{q_1, q_4\}$

Show how to construct a regular expression α equivalent to this automaton, i.e. $L(\alpha) = L(M)$.



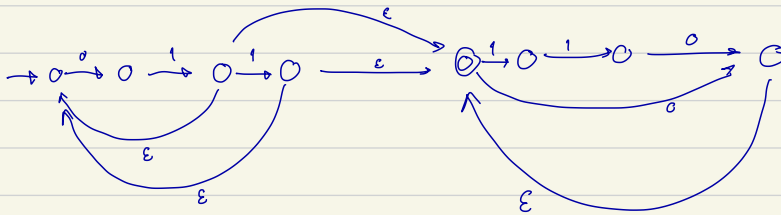
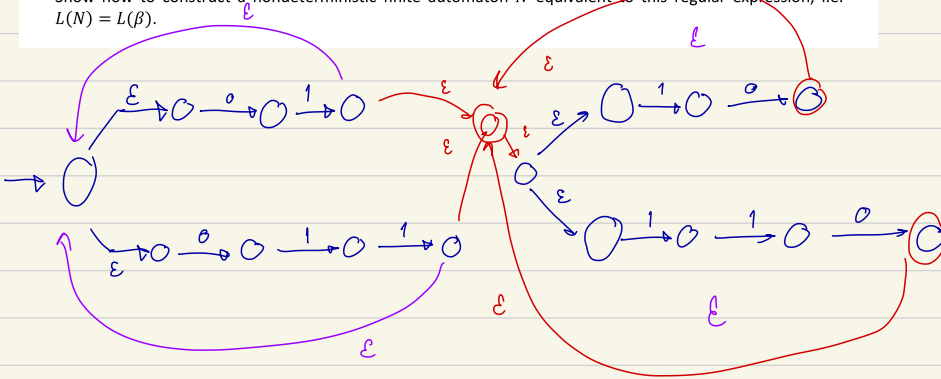
$$0(0101)^* \cup (0(0101)^*00 \cup 10^*1) \cup (11000)(0101)^*00$$

PROBLEM 2

Let β be the following regular expression over the alphabet $\Sigma = \{0,1\}$:

$$\beta = (01 \cup 011)^+(10 \cup 110)^*$$

Show how to construct a nondeterministic finite automaton N equivalent to this regular expression, i.e. $L(N) = L(\beta)$.



PROBLEM 3

Let R be the following language over the alphabet $\Sigma = \{0,1\}$: even number of state

$$R = L((0 \cup 10)^*) - L((\Sigma\Sigma)^*)$$

Find a regular expression φ such that $L(\varphi) = R$.

$\downarrow$
0, ε $\downarrow$
 $\varepsilon, 10, \dots$

$$(10 \cup 00)^* 0 (10 \cup 00)^*$$

$\Sigma \cdot \Sigma = \{00, 01, 10, 11\}$

PROBLEM 4

For each language below, prove or disprove whether it is regular. Here, $\Sigma = \{0,1\}$.

- (1) $L_1 = \{w \in \Sigma^* \mid w \text{ is a palindrome}\}$. Note that a *palindrome* is a string that reads the same forward and backward, such as 10101 and 0110.

1. Assume L_1 is regular

2. If L_1 is regular, L_1 should be able to use pumping lemma in every case.

assuming case $0^p 1 0^p$

$$\underbrace{0 \dots 0}_{x} 1 \underbrace{0 \dots 0}_z$$

$xyy \notin L_1$; contradiction.

L_1 is not regular

- (2) $L_2 = \{0^m 1^n \mid m \geq n \geq 1 \text{ and } m \text{ is divisible by } n\}$.

1. Assume L_2 is regular

2. L_2 needs to be pumpable in every case.

Assuming $0^q 1^p$

$$\underbrace{0 \dots 0}_q \underbrace{1 \dots 1}_p$$

$x = \varepsilon$

$xz \notin L_2$

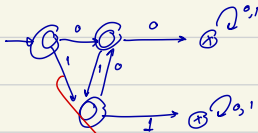
L_2 is not regular

(3) $L_3 = \{ww \mid w \in \Sigma^*\} \cap \{w \in \Sigma^* \mid w \text{ contains no consecutive 0s and no consecutive 1s}\}$.

1. Assume L_3 is regular.

2. If L_3 is regular, α_1 and α_2 needs to be regular.

3. α_2 is be able to draw in form of automaton.



4. If α_1 is regular, α_1 should be pumpable for every case.

$0^p 1 0^p$
 w

$0 \dots 0 \mid 1 \mid 0 \dots 0 1$
 $x \quad y \quad z$

$xyz \notin \alpha_2$

5. α_1 is not regular

L_3 is not regular

regular.

